ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

DATA COMMUNICATION AND NETWORKING

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Data Communication and Networking paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct, and other correct examples. Give method marks for correct working in calculations.

SECTION A

QUESTION 1

1.1 1.1.1 C ✓ 1.1.2 A ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 D ✓ 1.1.6 B ✓ 1.1.7 C ✓ 1.1.8 A ✓ 1.1.9 B ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 H ✓ 1.3.2 F ✓ 1.3.3 G ✓ 1.3.4 B ✓ 1.3.5 J ✓ 1.3.6 C ✓ 1.3.7 I ✓ 1.3.8 E ✓ 1.3.9 D ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 7 Application ✓; 6 Presentation ✓; 5 Session ✓; 4 Transport ✓; 3 Network ✓; 2 Data link ✓; 1 Physical ✓ (7)

2.2 Application ↔ OSI 5, 6, 7 ✓; Transport ↔ OSI 4 ✓; Internet ↔ OSI 3 ✓; Network access (link) ↔ OSI 1 and 2 ✓ (4)

2.3 Connection: TCP connection-oriented (three-way handshake) ✓ – UDP connectionless ✓; Reliability: TCP guarantees delivery, order and error checking with acknowledgements ✓ – UDP no guarantee, faster ✓; Example: TCP – web pages, e-mail, file transfer ✓ – UDP – video calls, online gaming, DNS, streaming ✓ (6)

2.4 (a) FTP – transfers files between a client and a server ✓ (b) DHCP – automatically assigns IP addresses, subnet mask, gateway and DNS settings ✓ (c) IMAP – retrieves e-mail while keeping it on the server so it syncs on several devices ✓ (3)

[20]

QUESTION 3

3.1 10101100 ✓ . 00010000 ✓ . 00101101 ✓ . 10000010 ✓ (4)

3.2.1 2 bits ($2^2 = 4$ subnets) ✓ (1)

3.2.2 255.255.255.192 ✓; /26 ✓ (2)

3.2.3 6 host bits: $2^6 - 2$ ✓ = 62 hosts ✓ (2)

3.2.4 Network 192.168.10.64 ✓; first host 192.168.10.65 ✓; last host 192.168.10.126 ✓; broadcast 192.168.10.127 ✓ (4)

3.3 Any TWO: far more addresses (128-bit) ✓; no need for NAT ✓; built-in security (IPsec); simpler header / more efficient routing; automatic address configuration (2)

[15]

QUESTION 4

4.1 Any FIVE with function (1 + 1 each): router – connects the LAN to the Internet / other networks ✓✓; switch – connects computers in the LAN and forwards frames by MAC address ✓✓; wireless access point – connects Wi-Fi devices ✓✓; firewall – filters traffic to protect the network ✓✓; fibre ONT / modem – converts the fibre signal for the router ✓✓; network interface cards – connect each computer; patch panel – organises cables; server – stores shared files (10)

4.2 (Extended) star topology ✓; reasons (any TWO): one cable failing does not affect the others ✓; easy to add devices / expand ✓; easy to troubleshoot; high performance with switches (3)

4.3 UTP is limited to 100 m ✓, so use fibre-optic cable between the floors, or add a switch / repeater in the middle ✓. (2)

4.4 Visitors could access confidential client and accounting files ✓; a separate network (guest VLAN / SSID) keeps staff data secure and limits malware spread ✓. (2)

4.5 Peer-to-peer: all computers are equal and share resources directly, no central server ✓; client–server: a central server provides services (files, security, backups) to clients ✓. Client–server, because 25 users need central security, backups and file sharing ✓. (3)

[20]

QUESTION 5

5.1 Any FIVE: strong passwords and user accounts with permissions ✓; firewall ✓; antivirus / anti-malware on all devices ✓; WPA2 / WPA3 encryption on Wi-Fi ✓; regular updates and patches ✓; backups; separate guest network; disable unused ports; VPN for remote access; staff training on phishing; physical security of the server room (5)

5.2 An APIPA address – the computer could not get an address from the DHCP server ✓✓. Any TWO steps: check that the network cable is connected / Wi-Fi is on ✓; run ipconfig /release and ipconfig /renew ✓; restart the router / check the DHCP server; set a static IP (4)

5.3 It shows the path (each router or hop) that packets take to the destination ✓ and the time taken at each hop, helping to find where a connection fails ✓. (2)

5.4 Symmetric encryption uses the same secret key to encrypt and decrypt ✓ – fast, but the key must be shared securely ✓; asymmetric encryption uses a public key to encrypt and a private key to decrypt ✓ – used e.g. in HTTPS / digital signatures, no need to share the private key ✓. (4)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE